A COMPANION TEXTBOOK

Scientific Writting

Principles and Practice

A five-chapter guide for medical students, from IMRAD structure to peer review

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ASIME

Academic Society for International Medical Education

Scientific Writing

Principles and Practice

A companion textbook for the five-class course

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Scientific Writing: Principles and Practice
A Companion Textbook

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The examples of studies, data and author names used in this book are illustrative and were invented for teaching purposes. They do not report real findings.

This book accompanies the course *Scientific Writing: Principles and Practice*, taught in the Honours Student Program, and its companion website, scientific-writing.vercel.app.

Preface

Every year I read theses and manuscripts by students who have done careful, original research and then struggled to put it on the page. The problem is almost never the science, and it is very rarely the English. It is that nobody has shown them how a scientific text is built: what each section is for, what a sentence should carry, and how a draft becomes a paper.

This book sets out to show exactly that. It accompanies a course of five classes that I teach to fourth-year medical students in the Honours Student Program, and it follows the same path. Chapter 1 explains what scientific writing is for and introduces the IMRAD structure. Chapter 2 works at the level of the sentence and the paragraph. Chapter 3 builds an introduction, from the opening line to the hypothesis. Chapter 4 covers results, figures, the discussion and citation. Chapter 5 deals with everything that happens after the first draft: editing, feedback, peer review and writing for wider audiences.

The book is deliberately short. It is meant to be read alongside the classes, kept open while you write, and returned to when a supervisor or reviewer sends your draft back. Its advice is practical and its examples are small, so that each principle can be seen in a single sentence or paragraph.

I have written it for students whose first language is not English as much as for those for whom it is. Clear scientific writing does not depend on a large vocabulary or elegant style. It depends on structure, precision and honesty, and all three can be learned.

Branko Aleksic

Nagoya, November 2026

How to use this book

Each chapter corresponds to one class and follows the same pattern, so that you always know where to look.

Learning outcomes
On the opening page of each chapter: what you should be able to do by the end.

Before and after examples
A weak version beside a stronger one, with a one-line explanation of the difference.

Key points and tips
Blue boxes summarize a principle; yellow boxes give practical advice.

Try it
Short exercises you can do alone or with a classmate, in a few minutes each.

Writing task
One short piece of writing per chapter, the same task as on the course website.

Review questions
Multiple-choice questions, with answers and explanations in Appendix B.

WEAK VERSION

"Examples that show a common problem appear in a box like this."

STRONGER VERSION

"The improved version appears beside it, like this."

> A short note under each pair says what changed and why.

The companion website

The course website carries every slide from the five classes, a quiz and a writing task for each class, and a discussion page where you can ask questions. If you type your writing tasks there, your instructor can read them.



COMPANION WEBSITE

scientific-writing.vercel.app

Scan the code or type the address. Log in with your name and a 4-digit PIN; the same name and PIN restore your progress on any device.

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1

CHAPTER ONE

Foundations of Scientific Writing

What scientific writing is for, and how a paper is built.

AFTER THIS CHAPTER YOU CAN

- Explain why scientific writing has its own rules, and why those rules exist
- Name the major types of scientific documents and what each is for
- Describe the IMRAD structure and what belongs in each section
- Recognize the parts and purpose of a title and an abstract

1.1

Why scientific writing matters

Scientific writing is a way of reporting research so that other people can understand it, judge it, and build on it. That definition is worth reading twice, because each of its three verbs places a demand on the writer. To be *understood*, the text must be clear. To be *judged*, it must show its evidence and its methods. To be *built on*, it must be precise enough that someone else can repeat the work or extend it.

It is tempting to think of scientific writing as ordinary English made more formal. It is not. Its conventions follow a logic that mirrors the scientific method itself, and every choice of structure, tense and word order exists to help a reader evaluate your evidence quickly and fairly. For an Honours student this is the skill that turns months of research into something the field can use. A finding that cannot be understood, checked or repeated has not really been reported at all.

Who is reading?

A paper has several audiences at once, and each of them wants something different from it.

Peer researchers
Need enough detail to replicate your method and trust your results.

Clinicians
Want to know one thing: does this change what I do for my patients?

Reviewers and editors
Judge rigor, novelty, and whether the claims match the evidence.

The public and policy makers
Rely on plain-language summaries to make real-world decisions.

No writer can address each of these readers separately. The solution the scientific community arrived at is a standard structure: when every paper is organised the same way, each reader knows where to find the part they need.

A different kind of writing

Many habits that serve you well in essays and creative writing work against you here. A good story may hold back its ending; a good paper states its main finding early and directly. In everyday writing a personal, varied style is a virtue; in science the voice is precise and consistent. And where a novelist may treasure ambiguity, a scientist treats it as a defect to be removed.

TABLE 1.1 Everyday writing and scientific writing compared

Everyday or creative writing	Scientific writing
Can build suspense and save the punchline	States the main finding early and directly
Voice and style are personal and varied	Voice is precise, consistent and impersonal
Structure is flexible, even surprising	Structure is conventional and predictable (IMRAD)
Ambiguity can be a feature	Ambiguity is a defect to be removed

When writing fails, it usually fails in a small number of familiar ways: unnecessary jargon or undefined abbreviations; sentences carrying two or three ideas at once; results and interpretation mixed together; no clear thread from the question to the conclusion; or a structure that does not match what the reader expects. Each chapter of this book addresses one or more of these failures.

Authorship is a responsibility, not a reward

Being listed as an author means taking public responsibility for the accuracy of the work. The most widely used international criteria, those of the International Committee of Medical Journal Editors (ICMJE), require all of the following: a substantial contribution to the work, drafting or critically revising the manuscript, approval of the final version, and agreement to be accountable for it. Collecting data alone, without intellectual input, is normally recognised in the acknowledgments rather than by authorship.

Accountability protects the scientific record. If an error or misconduct is found later, the authors are the ones held responsible. This discourages “gift authorship”, the practice of adding someone who did not meaningfully

contribute, and it protects genuine contributors, whose names are attached only to work they can vouch for.

> **PRACTICAL ADVICE**
>
> Discuss authorship with your supervisor early, before the paper is drafted. Agreeing on who will be an author, and in what order, is far easier at the start of a project than at the end.

1.2

Types of scientific documents

You will meet six kinds of document repeatedly during your training. They differ in length and purpose, but, as we shall see, they share an underlying logic.

TABLE 1.2 Common scientific documents

Document	Typical length	Primary purpose
Original article	3,000–5,000 words	Report new data and its interpretation
Review article	4,000–8,000 words	Synthesize and critically evaluate prior work
Case report	1,000–1,500 words	Document an instructive or novel clinical case
Conference abstract	250–400 words	Summarize work for presentation at a meeting
Thesis or dissertation	10,000+ words	Demonstrate independent, extended research
Grant proposal	Set by the funder	Argue persuasively for research not yet conducted

Choosing the right format is part of the skill. A student with preliminary pilot data and a ten-minute conference slot should write a conference abstract, which is short, quickly reviewed and designed for work in progress; a full original article would be premature.

This book concentrates on the original research article. It is the strictest and most complete expression of the logic that all these documents share, which makes it the best format in which to learn. Once you can structure an original

article, reviews, case reports and theses become much easier to organise.

The other parts of a manuscript

A complete manuscript contains more than its main text. The **title page** gives the title, authors, affiliations and corresponding author. The **acknowledgments** credit contributors who do not meet the criteria for authorship. The **funding statement** discloses who paid for the research and what role they played, and the **conflict of interest** statement discloses any financial or personal interests that could bias the work. The **references** list every source in the style the journal requires, and **supplementary material** holds additional data, methods or figures.

Journals insist on funding and conflict-of-interest disclosures because transparency builds trust. Disclosure does not mean that funded research is untrustworthy. It means that readers can judge for themselves whether a funding source might have shaped the design or the interpretation. An undisclosed conflict that comes to light later can seriously damage a researcher's credibility.

1.3

The IMRAD structure

Almost every research article is built on the same four-part frame, known by its initials as IMRAD: Introduction, Methods, Results and Discussion. Each part answers one question.

I	M	R	D
Introduction	**Methods**	**Results**	**Discussion**
Why did you do this study?	What did you do, exactly?	What did you find?	What does it mean?

IMRAD is not an arbitrary convention. It follows the scientific method step by step: a question, a procedure, an observation, an interpretation. That is why it has lasted, and why readers in every field can navigate a paper they have never seen before.

The Introduction: a funnel

The Introduction opens with the broad clinical or scientific context and explains why anyone should care. It then narrows to what is already known, citing the key prior work; identifies the specific gap, the thing that remains unknown or unresolved; and ends at its narrowest point with your precise aim, question or hypothesis. Chapter 3 builds a full introduction step by step.

The Methods: a recipe

The Methods section is written so that a competent researcher elsewhere could repeat your study exactly. It is always in the past tense, because you are reporting what you did, not instructing the reader. It covers participants or materials, procedures, measurements and statistical analysis. No results and no interpretation belong here.

The Results: just the findings

The Results section reports what was observed, in words supported by figures and tables. It follows a logical order, usually the order in which the methods were introduced, and it states numbers precisely, with exact values, units and appropriate statistics. The single most common student error is to let interpretation slip in here.

MIXES IN INTERPRETATION

"The treatment group showed lower blood pressure, which proves the drug is effective and should be adopted as first-line therapy."

STAYS OBJECTIVE

"Mean systolic blood pressure was 8 mmHg lower in the treatment group than the control group ($p<0.01$)."

> "Proves" and "should be adopted" are interpretive claims. They belong in the Discussion, supported by comparison with other evidence.

The Discussion: what it means

The Discussion opens by restating the main finding in plain terms. It compares your results with previous studies and explains agreement or disagreement, states the limitations honestly, and closes with the practical or scientific

significance of the work. Limitations belong here, before the conclusion, because they are a judgement about how much weight the findings can bear, and judgement is the business of the Discussion.

RESULTS OR DISCUSSION?

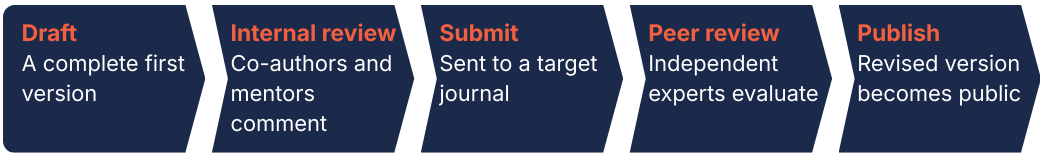
Consider the sentence *"Blood pressure was significantly lower in the treatment group ($p<0.01$)."* It reports an observed finding with a statistic and makes no comparison with other studies, so it belongs in the Results. A Discussion version might read: *"This reduction suggests the treatment improves cardiovascular risk, consistent with prior trials."* Results say what happened. Discussion says what it means.

TABLE 1.3 IMRAD at a glance

Section	Guiding question	Typical verb tense
Introduction	Why did you do this study?	Present for established facts; past for prior studies
Methods	What did you do, exactly?	Past tense throughout
Results	What did you find?	Past tense throughout
Discussion	What does it mean?	Present for interpretation; past for your findings

From draft to publication

Writing the manuscript is one stage in a longer journey, which Chapter 5 follows in detail.



1.4

Title, abstract and keywords

The title: your paper's only guaranteed reader

Most people who encounter your paper will read only its title. It must therefore be informative enough to stand alone. Where possible it should describe the population, the intervention or exposure, and the key outcome, and it should

contain every important term a colleague might type into a search engine. Vague or clever titles fail on both counts.

TOO VAGUE

"A *Study of Blood Pressure in Patients*"

SPECIFIC AND SEARCHABLE

"*Effect of an 8-Week Walking Program on Systolic Blood Pressure in Adults with Stage 1 Hypertension*"

> The strong version names the intervention, its duration, the population and the outcome. A reader, or a search engine, knows exactly what is inside.

The abstract: the paper in miniature

The abstract is a self-contained summary, usually of 200 to 300 words and subject to a strict journal limit. It mirrors IMRAD in miniature: background, aim, methods, key results and conclusion. It must be understandable on its own, because databases, search engines and busy readers often see nothing else. For that reason it is usually written last, even though it appears first.

Abstracts come in two forms. A **structured** abstract uses labelled subheadings such as Background, Methods, Results and Conclusion; it is common in clinical and biomedical journals and is easy to scan. An **unstructured** abstract is a single continuous paragraph, more common in basic science, and needs stronger internal logic to guide the reader.

GENERIC OPENING

"*This study looks at an important health topic and presents some interesting findings.*"

SPECIFIC AND INFORMATIVE

"*Background: Adherence to hypertension medication remains low among young adults, though the reasons are poorly understood.*"

> A strong opening states the specific topic and why it matters. It never offers a vague preview of "something interesting".

DRAFTING AN ABSTRACT EFFICIENTLY

- Draft it last, but budget real time for it. It is often the most-read part of your paper.
- Write one sentence per IMRAD section first, then expand within the word limit.
- Check the target journal's word limit and required structure before you finalise.
- Read it aloud, on its own. Does it still make sense without the paper?

Keywords: making your work findable

Ask yourself what you would type into a database to find your own study. Those words should appear in the title, in the abstract and in the keyword list, repeated naturally. Use standard terms in place of abbreviations and laboratory jargon. Think like your future reader, who does not yet know the project, and not like the person who has lived with it for a year.

KEY TAKEAWAYS

- ✔ Scientific writing exists to help readers evaluate evidence quickly and fairly.
- ✔ Different documents (article, review, case report, thesis) serve different purposes.
- ✔ IMRAD is the shared logic behind almost all of them: question, method, finding, meaning.
- ✔ The title and abstract carry outsized weight, because most readers never go further.

1.5

Practice

TRY IT

1. Think of a paper or report you found hard to read. Write down exactly what made it difficult, and match each difficulty to one of the five common failures listed in section 1.1.
2. Write three keywords for a study you might carry out. Would a stranger searching for your topic use the same words?
3. Take the title of a paper you have read recently. Does it name the population, the intervention or exposure, and the outcome? Rewrite it so that it does.

WRITING TASK 1

Summarize a study in four sentences

Pick a real study you know, or invent a simple one, and summarize it in exactly four sentences, one for each IMRAD section. Each sentence should answer that section's guiding question: why the study was done, what was done, what was found, and what it means.

Check yourself: Does each sentence stay in its own section, with no interpretation in the Results sentence and no findings in the Methods sentence? Could a reader tell the four sentences apart without labels?

> *Good scientific writing doesn't decorate an idea — it gets out of its way.*

Review questions

1.1 Which of these is NOT a core value of scientific writing?

A. Suspense — withholding the main finding until the end

B. Reproducibility — enough detail for others to repeat the work

C. Clarity — one idea communicated without confusion

D. Objectivity — separating observation from interpretation

1.2 A student wants to present preliminary pilot data at a 10-minute conference slot. Which format fits best?

A. A full original article

B. A grant proposal

C. A thesis chapter

D. A conference abstract

1.3 Where should a study's limitations be discussed?

A. In the Methods, as a disclaimer

B. In the Results, after the main findings

C. In the Discussion, before the conclusion

D. In the Abstract only

1.4 An abstract should generally be understandable:

A. Only by co-authors who know the project

B. On its own, without reading the rest of the paper

C. Only after reading the full paper

D. Only by statisticians

2

CHAPTER TWO

Writing Clear and Accurate Sentences

Sentence structure, voice, and cutting the words that don't earn their place.

AFTER THIS CHAPTER YOU CAN

- Build sentences that carry one idea each
- Choose active or passive voice deliberately
- Find and cut redundant phrases and vague quantifiers
- Organise a paragraph around a single topic sentence

2.1

One idea per sentence

Consider this sentence: "*It was found that the levels of the biomarker, which had been previously reported in several other studies conducted in different populations, were significantly elevated.*" Nothing in it is ungrammatical, yet the reader must hold a great deal in mind before reaching the point. It buries the finding (the biomarker was elevated) beneath a second idea (others have reported it before), and it opens with three words, "It was found", that say nothing.

Clear scientific writers share four habits, and this chapter takes them in turn.

One idea, one sentence
Never let a sentence carry more than one claim.

Deliberate voice
Choose active or passive according to what matters.

Cut redundancy
Every surviving word should earn its place.

One purpose, one paragraph
Build each paragraph around a single topic sentence.

The first habit is the foundation of the rest. Every sentence should make exactly one claim that the reader can check off before moving on. If a sentence needs "and", "which" or "while" to hold two unrelated facts together, consider splitting it. Short does not mean simplistic: the goal is precision, not brevity for its own sake, and a long sentence is perfectly appropriate when its ideas are genuinely and logically connected.

TWO IDEAS COLLIDING

"*The intervention reduced HbA1c by 0.8%, and adherence was measured using a validated questionnaire administered at baseline and follow-up.*"

SPLIT INTO TWO CLAIMS

"*The intervention reduced HbA1c by 0.8%. Adherence was measured using a validated questionnaire at baseline and follow-up.*"

> The result and the measurement method are two separate claims. Each deserves its own sentence.

Nominalizations: verbs disguised as nouns

A nominalization turns a strong verb into an abstract noun: "investigate" becomes "conduct an investigation of". The change adds words and moves the sentence away from a clear actor and a clear action. Scan your drafts for the endings *-tion*, *-ment* and *-ance*; each is a candidate for a stronger verb. Not every nominalization is wrong, but most sentences read better with the verb restored.

BURIED VERB

"The research team conducted an investigation of the relationship between diet and inflammation."

VERB RESTORED

"The research team investigated the relationship between diet and inflammation."

> The two versions mean exactly the same thing. The second says it in three fewer words.

2.2

Active and passive voice

Students are often told that scientific writing must be passive, or that it must never be. Neither rule is right. Active and passive are two tools for two jobs.

TABLE 2.1 Two voices, two jobs

Active voice	Passive voice
"We measured blood pressure every 4 weeks."	"Blood pressure was measured every 4 weeks."
Shorter, more direct, easier to follow	Shifts the focus onto the action or object
Best when the actor matters	Best when the actor is obvious, irrelevant or unknown

Use the active voice by default: it is shorter and puts the responsible agent at the front. Use the passive deliberately, most often in the Methods, where "Samples were centrifuged" is entirely adequate because the reader does not need to know who operated the centrifuge. Avoid the passive when it hides who is responsible for a judgement. "It is believed that…" conceals whether the belief is yours or someone else's. Many journals now prefer the active voice throughout, so check the author guidelines.

WHO DECIDED THIS?

"It was decided that the control group would not receive follow-up assessment."

CLEAR RESPONSIBILITY

"The research team excluded the control group from follow-up assessment to reduce participant burden."

> The revision restores the actor and adds the missing reason. Both are information the reader needs.

TABLE 2.2 Typical voice in each IMRAD section

Section	Typical voice	Why
Introduction	Mostly active	You are making claims and setting up an argument
Methods	Mixed; passive common	The actor is assumed; the procedure matters more
Results	Mostly active	A direct statement of what was found reads more clearly
Discussion	Mostly active	You are the one interpreting, so own that role

The test is always the same: does the actor matter? In "*The questionnaire was completed by participants after each session*" the actors are named anyway, so the active version, "*Participants completed the questionnaire after each session*", is shorter and loses nothing. Had the sentence been about the questionnaire and its timing, the passive would have been a reasonable choice.

Hedging: saying how sure you are

Hedges such as "may", "could", "suggests" and "appears to" calibrate how confident a claim is. Too little hedging overstates certainty; too much makes every sentence sound tentative and weak. Match the hedge to the evidence: strong data support strong language, and preliminary data need softer claims. Avoid stacking hedges. "May possibly suggest" hedges the same idea three times.

2.3

Cutting redundancy

Redundant phrases feel formal but add no information. Every extra word costs the reader attention, and journals enforce strict word limits, so redundancy is the easiest place to find space. Cutting it is not about writing less. It is about writing exactly enough.

TABLE 2.3 Redundant phrases and their fixes

Wordy	Concise
due to the fact that	because
in order to	to
a total of 42 patients	42 patients
it is important to note that	(usually, just delete it)
at this point in time	now; currently
the majority of	most

PADDED

"Due to the fact that a total of 42 patients were enrolled, it is important to note that the sample size was adequate in order to detect a clinically meaningful difference."

TRIMMED

"Because 42 patients were enrolled, the sample size was adequate to detect a clinically meaningful difference."

> The same information in roughly 30% fewer words, and easier to read in one pass.

Vague quantifiers undermine precision

Words such as "significant", "many", "several" and "a large proportion" feel scientific but carry no exact meaning. "Significant" is especially dangerous, because it has a strict statistical meaning that differs from everyday usage. Whenever you have the actual number, use it. Reserve "significant" for statistical significance, and use "substantial" or "considerable" when you mean size in a general sense.

VAGUE

"A large proportion of participants reported significant improvement in symptoms."

PRECISE

"78% of participants (39/50) reported symptom improvement, and the difference from baseline was statistically significant ($p<0.01$)."

> The revision separates the proportion (78%) from the statistical claim ($p<0.01$). "Significant" is no longer doing double duty.

2.4

Building clear paragraphs

The principle that governs sentences also governs paragraphs. A paragraph is a group of sentences serving one clear purpose. It starts with a topic sentence that states its main point, and every following sentence supports, explains or provides evidence for that point. A sentence that does not relate to the topic sentence probably belongs in a different paragraph.



If you read a paragraph and cannot tell what its main point is after the first sentence, the most likely cause is a missing topic sentence.

WANDERING

"Participants completed the survey online. Some used mobile devices. The response rate varied by age group. Data were checked for completeness before analysis."

ANCHORED BY A TOPIC SENTENCE

"Survey completion varied by device and age group. Most participants used mobile devices, and response rates were highest among younger age groups. All responses were checked for completeness before analysis."

> The revision opens with the point (completion varied), so every following sentence has a clear job: supporting that claim.

Transitions between paragraphs

A transition tells the reader how the next paragraph relates to the one before: whether it builds on a point, contrasts with it, or narrows from the general to the specific. Connector words such as “Furthermore” and “Moreover” are no substitute for a real logical link. The strongest transitions echo a key word or idea from the last sentence of the previous paragraph.

MECHANICAL ONLY

“*...supports this mechanism. Furthermore, other factors may also be relevant.*”

LOGICALLY CONNECTED

“*...supports this mechanism. This same pathway may also explain the age-related differences described below.*”

> The second version links specific content across paragraphs, not just a generic connector word.

SENTENCE-LEVEL SELF-CHECK

- **One idea?** Does each sentence make exactly one claim?
- **Right voice?** Is active or passive chosen deliberately, not by habit?
- **No padding?** Has every redundant phrase been cut?
- **Clear topic sentence?** Does every paragraph open with its main point?

KEY TAKEAWAYS

- ✓ Give each sentence one idea; split sentences that carry two.
- ✓ Choose active or passive voice deliberately, according to whether the actor matters.
- ✓ Hunt down and cut redundant phrases; replace vague quantifiers with numbers.
- ✓ Paragraphs need a clear topic sentence and a single unifying purpose.

2.5

Practice

TRY IT

1. Rewrite the biomarker sentence that opens this chapter so that the finding comes first. What did you cut?
2. Find three nominalizations in something you have written and restore the verbs.
3. Take any paragraph of your own and underline its topic sentence. If you cannot find one, write it.

WRITING TASK 2

Untangle an overloaded sentence

Rewrite the following as two or three clear, active-voice sentences with no redundant phrases: *"A total of 60 participants were recruited for the study and blood samples were collected at three time points and the results were subsequently analyzed using a mixed-effects model, which is important to note given the repeated-measures design."*

Check yourself: Did you split the separate claims into their own sentences? Is every verb doing real work? Did you cut throat-clearing phrases such as "it is important to note that"?

> *If a sentence needs to be read twice to be understood, it needs to be rewritten once.*

Review questions

2.1 A sentence containing three linked clauses joined by 'and' is usually a sign that you should:

A. Add more clauses for completeness

B. Move it to the abstract

C. Convert it to passive voice

D. Consider splitting it into separate sentences

2.2 In a Methods section, passive voice is most defensible when:

A. The actor performing the action is obvious or unimportant

B. You want to sound more formal

C. You are unsure who did the work

D. You need to lengthen the section

2.3 A single small pilot study found a promising trend that didn't reach statistical significance. Which claim best matches the evidence?

A. "This treatment cures the condition."

B. "This treatment may offer modest benefit, though further study is needed."

C. "This treatment has no effect whatsoever."

D. "This treatment possibly could perhaps help somewhat."

2.4 Which revision best fixes "a total of 15 subjects were recruited for participation in the study"?

A. "15 subjects were recruited."

B. "A total of 15 subjects participated."

C. "Subjects, a total of 15, were recruited for the study."

D. No change needed

2.5 A paragraph with no topic sentence and several unrelated facts is best fixed by:

A. Adding a strong final sentence to summarize

B. Converting every sentence to passive voice

C. Adding a topic sentence and/or splitting it into focused paragraphs

D. Making all sentences the same length

3

CHAPTER THREE

Building the Introduction and Literature Review

From broad context to a precise, defensible research question.

AFTER THIS CHAPTER YOU CAN

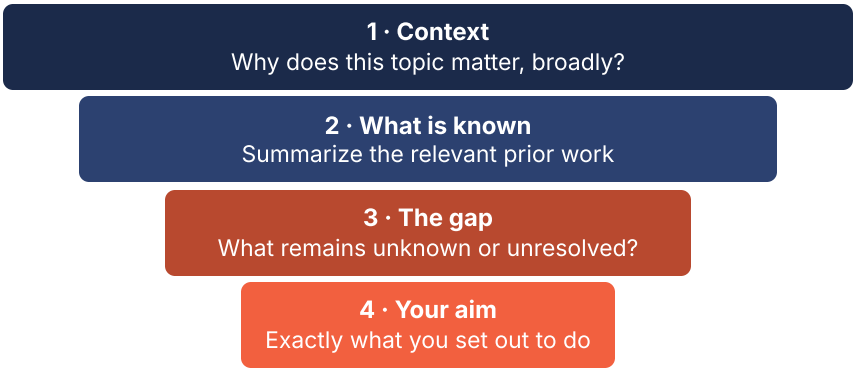
- Structure an introduction as four narrowing moves
- Synthesize prior studies instead of listing them
- State a specific research gap that your study can close
- Write a testable aim, question or hypothesis

3.1

The introduction as a funnel

The introduction is the first section a reader meets and, for most experienced writers, one of the last they draft. It is easier to justify a study you have completed than to predict how you will frame one, so many authors write the Methods and Results first and return to the Introduction once they know exactly what they have to introduce. If your first attempt feels weak, that is normal and not a sign of failure. This chapter builds the section piece by piece.

A good introduction makes four moves, each narrower than the last. Together they form a funnel that carries the reader from the wide world to your specific study.



Move 1: establish why it matters

Start with the clinical or scientific importance of the general topic, not with a dictionary definition and not with “Since ancient times…”. Use the opening two or three sentences to orient any reader, including one from outside your subfield. Ground the importance in real consequences: disease burden, unresolved clinical questions, cost.

TOO BROAD, NO STAKES

"Diabetes has been studied by scientists for many years and is an important disease that affects many people around the world."

SPECIFIC AND CONSEQUENTIAL

"Type 2 diabetes affects over 400 million adults worldwide and remains a leading cause of preventable cardiovascular disease."

The stronger version names a real number and a specific consequence. It earns the reader's attention instead of assuming it.

Pacing the four moves

As a rough guide for a typical article, context takes 10–15% of the introduction, what is known 40–50%, the gap 15–20% and the aim 10–15%. A common student mistake is to spend too many paragraphs on background and then rush the gap and the aim. If a reader reaches the end of your introduction still unsure what the study tested, the pacing is the likely cause. An introduction longer than about one printed page usually contains material that belongs in the Discussion. The aim should almost always be the final sentence or two.

3.2

Reviewing the literature

The second move summarizes what is already known. The goal is **synthesis, not a list**: group studies by theme or finding, not one sentence per paper. Cite the studies most directly relevant to your question, not everything you have read. Note where studies agree, where they conflict, and what methods they used. Every citation should do work, advancing the argument toward your gap.

TABLE 3.1 Listing sources and synthesizing them

Listing (weak)	Synthesizing (strong)
"Smith et al. found X. Jones et al. found Y. Lee et al. found Z."	"Several studies report X (Smith; Jones), though findings on Y remain inconsistent (Lee vs. Chan)."
Reads like an annotated bibliography	Groups studies by theme and highlights agreement or conflict
Leaves the reader to draw the connections	Does the interpretive work for the reader

A passage such as "*Reference [12] studied* X *in mice. Reference [13] studied a related question in rats. Reference [14] used a different method entirely*" shows the underlying problem clearly. The studies are reported one after another, but nothing says how they relate to each other or to the question at hand.

Primary and secondary sources

Primary sources are original research articles reporting new data. They are the gold standard for supporting a specific claim, and you should always prefer them when citing a specific finding. **Secondary sources**, such as review articles and textbooks, are useful for background and orientation but risky to cite for specific numbers. When a review gives you a figure, trace it back to the primary study and cite that.

Finding and managing sources

- Start with a focused search in PubMed or another relevant database, using specific terms and not single broad keywords.
- Prioritize recent, high-quality, peer-reviewed sources; include older foundational work only when it is needed.
- Use reference management software such as Zotero, EndNote or Mendeley from the very first source you save.
- Record why each source matters as you save it. You will have forgotten by the time you write.

A practical tool: the synthesis matrix

Before writing, sort your sources into a simple table organised by theme. The matrix makes conflicts and gaps visible before you write a single sentence, and it prevents the "list of studies" problem because it forces you to compare and not merely record. Update it as you read; it becomes the skeleton of your literature review.

TABLE 3.2 A simple synthesis matrix

Theme	Supporting studies	Agreement or conflict
Effect in adults	Smith (2021), Jones (2022)	Consistent positive effect
Effect in adolescents	Lee (2020)	Single study; unconfirmed
Optimal duration	Chan (2019), Park (2023)	Conflicting: 4 vs. 8 weeks

Read the right-hand column again. “Single study; unconfirmed” and “Conflicting” are not just notes about the literature. They are candidate gaps.

3.3

Identifying the gap

The gap is the hinge of the introduction: the specific thing that remains unknown, untested or unresolved after your review of the literature. A strong gap statement often begins with “However”, “Despite this” or “It remains unclear whether…”. It must follow logically from what you have just summarized and cannot come out of nowhere. A vague gap, such as “more research is needed”, is not a gap. It is an admission that you have not found one yet.

Population gap
Studied in one group (for example adults) but not in another (adolescents).

Methodological gap
Earlier studies used weak designs; a stronger method has not been applied.

Conflicting evidence
Existing studies disagree, and the reason is unclear.

Contextual gap
Findings from one setting, country or era may not generalize.

NOT ACTUALLY A GAP

“Further research is needed to better understand this topic.”

A GAP YOU CAN ACT ON

“No prior study has examined this intervention in adolescents, despite its established efficacy in adults.”

> The strong version names exactly who has not been studied and why that matters. A reader understands at once what your study will add.

Move 4: the aim

The aim is the narrowest point of the funnel. State your specific aim, research question or hypothesis in direct, unambiguous language, using an opener such as "This study aimed to…", "We hypothesized that…" or "We tested whether…". The aim should map directly onto the gap you have just identified, with no surprises, and it is usually the very last sentence of the introduction.

PICO: structuring a clinical question

For clinical and epidemiological questions, the PICO framework forces precision on four elements at once: the **P**opulation, the **I**ntervention or exposure, the **C**omparison and the **O**utcome.

TABLE 3.3 PICO in practice

Element	Ask	Example study
Population	Who is being studied? Age, condition, setting	Adults aged 18–40 with newly diagnosed stage 1 hypertension
Intervention	What is being tested, given or observed?	8-week structured walking program (30 min, 5 times a week)
Comparison	Compared with what?	Usual care with no structured exercise program
Outcome	What is measured, and when?	Change in systolic blood pressure at 8 weeks

Assembled into a sentence, the four rows give a complete aim: *to determine whether an 8-week structured walking program, compared with usual care, lowers systolic blood pressure in adults aged 18–40 with newly diagnosed stage 1 hypertension.*

3.4

Research questions and hypotheses

The final sentence of an introduction can take three forms, and the choice depends on how much you are able to predict.

TABLE 3.4 Question, hypothesis or aim?

Form	When to use	Example
Research question	Exploratory or descriptive work	"How does sleep duration relate to exam performance?"
Hypothesis	You can predict a specific direction	"We hypothesized that shorter sleep duration is associated with lower exam scores."
Aim statement	Broader studies with several objectives	"This study aimed to characterize the relationship between sleep and academic outcomes."

Whichever form you choose, it must be testable. A testable aim names the variables, the population and the relationship being examined, so that anyone could design a study to answer it.

NOT TESTABLE

"This study aimed to explore issues related to sleep and academic performance."

TESTABLE AND SPECIFIC

"This study aimed to determine whether average nightly sleep duration predicts final exam scores among second-year students."

A hypothesis needs one thing more: a direction. It should name the groups being compared and the effect expected, so that the data can clearly support or refute it.

IMPRECISE

"We hypothesized that sleep affects academic performance."

PRECISE AND TESTABLE

"We hypothesized that students averaging fewer than 6 hours of sleep would score significantly lower on final exams than those averaging 7–8 hours."

KEY TAKEAWAYS

- An introduction is a funnel: context, what is known, the gap, your aim.
- A literature review synthesizes and groups sources; it does not just list them.
- A real gap is specific, follows from the evidence, and can be closed by your study.
- Your aim or hypothesis should map directly onto the gap you identified.

3.5

Practice

TRY IT

1. Write an opening sentence for a topic you care about that contains one number and one consequence.
2. Build a three-row synthesis matrix from papers you have read. Which row suggests a gap, and of which of the four kinds?
3. Write your own research question in PICO form, then join the four elements into a single aim sentence.

WRITING TASK 3

Write a testable aim and hypothesis

Write a testable research aim (one sentence) and a precise hypothesis (one sentence) for a topic of your choice. Name the population, the variables and, for the hypothesis, the expected direction of the effect.

Check yourself: Could someone else design a study from your aim alone? Does your hypothesis name a direction (higher or lower, more or less), and not just "a relationship"?

> *An introduction should make the reader feel the gap before you ever state your aim.*

Review questions

3.1 An introduction that opens with 'Since ancient times, disease has affected humanity...' is weak mainly because it:

A. Is grammatically incorrect

B. Is too broad to establish specific, concrete stakes

C. Uses passive voice

D. Is too short

3.2 The primary goal of the literature review section is to:

A. List every paper you read during your research

B. Summarize your own results in advance

C. Prove you read more papers than other researchers

D. Synthesize prior findings to set up your specific gap

3.3 You want to cite a specific statistic (e.g., a prevalence rate). What should you do?

A. Trace the number back to the primary study and cite that

B. Cite whichever source mentions the number, even a textbook

C. Avoid using specific numbers in the introduction

D. Cite a review article that references the number

3.4 A well-written research aim should:

A. Introduce a new topic not covered earlier in the introduction

B. Be broad enough to allow multiple different studies

C. Map directly onto the gap identified just before it

D. Be saved for the Discussion section instead

4

CHAPTER FOUR

Presenting Results and Discussion

Reporting findings objectively, then interpreting them honestly.

AFTER THIS CHAPTER YOU CAN

- Report a result completely, without interpreting it
- Choose between a figure and a table, and caption either so it stands alone
- Build a Discussion in four moves, with honest limitations
- Cite and paraphrase in a way that avoids plagiarism

4.1

Reporting results objectively

The Results and the Discussion are separate sections for a reason. In the first you report what you observed; in the second you say what you think it means. Keeping the two apart lets a reader accept your data while disagreeing with your interpretation, which is exactly the freedom a critical reader needs.

The core rule is simple: **results report; they do not interpret.** State what was observed, with numbers, directions and statistical significance. Save every "this suggests", "this proves" and "this means" for the Discussion.

INTERPRETS TOO EARLY

"The treatment group had fewer complications, demonstrating that the new protocol is safer and should replace the current standard of care."

STAYS OBJECTIVE

"Complications occurred in 4 of 50 patients (8%) in the treatment group, compared with 12 of 50 (24%) in the control group."

> "Demonstrating" and "should replace" are judgements. The Results section reports the numbers and lets the Discussion make the case.

By the same test, a sentence such as "*Mean recovery time was 3.2 days shorter in the intervention group* (95% CI: 1.1–5.3, p=0.004)" is entirely appropriate for the Results. It gives a value, its precision and a test, and it draws no conclusion.

What a complete result statement includes

Every claim in the Results needs a number attached, and one number is rarely enough.

TABLE 4.1 Elements of a complete result statement

Element	Example	Why it is needed
Exact value	"mean = 4.2 mmol/L"	Lets readers judge the actual magnitude
Variability	"SD = 0.8" or "95% CI: 3.9–4.5"	Shows precision, not just a point estimate
Sample size	"n = 42"	Needed to judge statistical power
Test and p-value	"p = 0.03, paired t-test"	Lets readers verify that the analysis was appropriate

Null and negative results

A result showing no significant difference is still a result, and it deserves the same rigor as a positive one. Never omit an outcome because it failed to reach significance; doing so distorts the scientific record. State the actual finding, for example "No significant difference was observed (p=0.34)", and do not imply that nothing happened. Reporting only the favorable results is a serious breach of research integrity.

Ordering the section

Most clinical studies open their Results with "Table 1", a comparison of baseline characteristics between groups such as age, sex and other variables relevant to the outcome. It lets readers judge whether the groups were comparable. Most modern guidelines advise against attaching p-values to these baseline differences; describe them in words.

After the baseline table, lead with the primary outcome, the result your study was designed to produce. Follow with the secondary outcomes in the order in which the Methods introduced them. Group related findings together, and use subheadings if the section is long.

4.2

Figures, tables and captions

Use a **figure** to show trends, patterns, distributions or comparisons visually. Use a **table** to present precise numbers that a reader may want to look up or compare directly. Never present the same data in both: choose the format that

serves the point. In either case the reader should be able to understand the display before returning to your main text.

A caption that stands alone

Many readers look at the figures before they read a word of the text, so a caption must make sense on its own. A reliable structure is: what is shown, then the key methodological detail, then how to read any symbols or error bars. Define every abbreviation used in the figure, even if it is defined elsewhere, and state the sample size and any statistical test where relevant.

CANNOT STAND ALONE

"*Figure 1. Results of the study.*"

SELF-CONTAINED

"*Figure 1. Mean systolic blood pressure (mmHg) at baseline and 12 weeks by treatment group (n=50 per group). Error bars represent 95% confidence intervals; *p<0.05 vs. control.*"

> The strong caption tells the reader what is plotted, the sample size and how to interpret the symbols. There is no need to hunt through the text.

Making figures easy to read, and honest

Legible labels
Axis labels and units large enough to read at presentation size.

Purposeful color
Color encodes meaning, not decoration, and still works in grayscale.

No clutter
Remove gridlines, 3D effects and anything that adds no information.

Consistent scale
Axis ranges that represent the data honestly and do not exaggerate effects.

The last of these deserves particular care. A bar chart whose y-axis starts at 90% instead of 0% makes a two-point difference look like a dramatic gap; the same chart with a full axis shows the true, modest size of the difference. Truncated axes are among the most common ways in which figures mislead, whether by accident or by design. As a careful reader, always check where the axis starts and judge the effect from the numbers, not from the height of the bars.

4.3

Building the Discussion

A strong Discussion makes four moves, and in a sense it runs the funnel of the Introduction in reverse: it starts from your specific finding and widens toward its meaning for the field.

1 · Restate The main finding, in plain language

2 · Compare With prior literature

3 · Limit Honest, specific limitations

4 · Signify Why the finding matters

Comparing with prior work, honestly

State clearly whether your finding agrees or disagrees with previous studies. When it conflicts with earlier work, propose a plausible reason, such as a different population, method or dose; do not simply note the conflict. Avoid cherry-picking only the studies that support your conclusion. This is where the literature review of Chapter 3 pays off, because you already know the landscape.

What counts as a real limitation

Sample
Small size, a single site, or a population that limits generalizability.

Measurement
Imperfect tools, self-report bias or missing data.

Design
Observational design, no control group or short follow-up.

Confounding
Variables you could not fully control or measure.

TOO VAGUE TO BE USEFUL

"This study has some limitations that should be considered."

SPECIFIC AND HONEST

"This single-center study enrolled only 45 participants, which may limit generalizability to broader populations."

> A vague limitation signals that the author has not seriously considered the study's weaknesses. Reviewers notice this immediately.

A good limitation statement names the weakness and says how it may have affected the result. Stated plainly, a limitation shows that you understand your own study, and readers trust authors who do.

Closing with significance

End the Discussion by answering three questions. What is the **clinical relevance**: how might this change practice, if it holds up? What is the **scientific relevance**: what does it add to the theoretical picture? And what are the **future directions**: which specific next study would extend the work? Keep each claim in proportion to the evidence. One small study rarely justifies a change in practice.

4.4

Citation and avoiding plagiarism

Citations let readers verify your claims and trace the evidence you built on. Under-citing implies that an idea is yours when it is not, and even when unintentional this is a serious matter. Over-citing common knowledge clutters the text and dilutes the citations that matter. Follow your target journal's citation style consistently from the first draft.

Direct copying
Using another author's exact words without quotation marks and a citation.

Patchwriting
Swapping a few words in someone else's sentence while keeping its structure.

Self-plagiarism
Reusing your own previously published text without disclosure.

Patchwriting is the form that students most often commit without realising it. Suppose the original reads: "*Exercise reduces the risk of cardiovascular disease in older adults.*"

PATCHWRITING

"*Physical activity lowers the risk of heart disease in elderly people.*"

GENUINE PARAPHRASE, WITH CITATION

"*Regular physical activity is a well-established protective factor against cardiovascular disease in aging populations (Author, Year).*"

> Patchwriting keeps the original sentence structure with synonyms swapped in, and it is still plagiarism. A real paraphrase reconstructs the idea in your own sentence and cites the source.

GOOD CITATION HABITS

- Save the full citation the moment you read a source. Do not rely on finding it again later.
- Use reference management software consistently, so that in-text citations and the reference list stay synchronized.
- Double-check every automatically generated citation; software occasionally introduces errors.
- Take notes in your own words, with the source attached, and note why each source matters.

KEY TAKEAWAYS

- ✔ Results report what was observed, with complete numbers; interpretation waits for the Discussion.
- ✔ Figures and tables, with their captions, should stand alone without the main text.
- ✔ A strong Discussion restates, compares, honestly limits, and signifies your findings.
- ✔ Cite carefully: genuine paraphrase and clear attribution protect your credibility.

4.5

Practice

TRY IT

1. Find a Results sentence in a published paper and check it against Table 4.1. Which of the four elements are present?
2. Choose a figure from a paper you have read and cover the main text. Can you understand it from the caption alone? Rewrite the caption if not.
3. Paraphrase one sentence from a source without looking at it while you write, then compare. Did you keep the original structure?

WRITING TASK 4

Report a result, then state a limitation

Write one Results-appropriate sentence reporting a finding (numbers, no interpretation), then one honest, specific limitation statement for the same hypothetical study. Avoid vague phrases such as “some limitations exist”.

Check yourself: Does your Results sentence avoid interpretation words such as “suggests” or “shows that”? Does your limitation name a specific weakness and say how it might affect the finding?

> *A limitation stated plainly is a strength; a limitation hidden is a liability waiting to be found.*

Review questions

4.1 Which phrase is most likely to signal that a sentence has drifted from Results into Discussion?

A. “This suggests that older patients respond better”

B. “Mean age was 42.3 years (SD 8.1)”

C. “15 of 60 participants withdrew before follow-up”

D. “Figure 2 shows the distribution of scores”

4.2 The same dataset should generally be presented:

A. In both a figure and a table, for redundancy

B. Only in a table, never a figure

C. In whichever single format best serves the reader's need

D. Only as raw numbers in the text

4.3 Which of these is the strongest limitation statement?

A. “There may be some limitations to this work.”

B. “Our study, like all studies, is not perfect.”

C. “Limitations are discussed elsewhere.”

D. “Because participants self-reported diet, recall bias may have affected accuracy.”

4.4 A strong closing to a Discussion section should generally:

A. Introduce new results not mentioned earlier

B. State the finding's significance and plausible next steps

C. Repeat the abstract word for word

D. End abruptly after the limitations

4.5 Reusing a paragraph you wrote in a previous paper, without citation, in a new manuscript is:

A. Always acceptable since you wrote it

B. Required practice for consistency

C. Only a problem if someone else notices

D. A form of self-plagiarism that should be avoided or properly disclosed

5

CHAPTER FIVE

Editing, Peer Review, and Beyond

Revising your own work, responding to reviewers, and reaching wider audiences.

AFTER THIS CHAPTER YOU CAN

- Edit a draft in three ordered passes
- Give feedback that is specific, and receive it without defensiveness
- Describe journal peer review and respond professionally to reviewers
- Translate a finding for readers outside your field

5.1

Self-editing your draft

No one writes a good paper in one attempt. The difference between an experienced writer and a beginner is rarely the quality of the first draft; it is the number and honesty of the revisions that follow. This chapter is about what happens after the draft exists: how to edit it yourself, how to use other people's comments, and what to expect once it is submitted.

Begin by separating two activities that are often confused. **Proofreading** catches typos and grammatical errors, and it is the very last step. **Editing** questions structure, logic and clarity; it happens much earlier, and repeatedly. Never edit and draft at the same time. Finish a rough draft first, then revise in passes, and expect to revise many times.

Three passes, in order

Work from the top down. There is no point polishing the sentences of a paragraph that the first pass will delete.

TABLE 5.1 A three-pass self-editing checklist

Pass	Ask yourself	Tool
1 · Structure	Does each section do its IMRAD job, in order?	Reverse outline
2 · Paragraph	Does each paragraph have one clear topic sentence?	Topic-sentence test
3 · Sentence	Is every sentence active, precise and non-redundant?	Read-aloud pass

Tricks that reveal hidden problems

- **Read it aloud.** Awkward phrasing is much easier to hear than to see.
- **Set it aside for a day.** Distance makes your own errors visible again.
- **Read only the topic sentences** of every paragraph, in order. They should reveal the paper's argument on their own.
- **Reverse outline your draft.** Write down what each paragraph actually does, and compare it with what you intended.

The topic-sentence test is the quickest of these. Take a paragraph you wrote recently and read only its first sentence. Does it predict what follows? If it does not, either the paragraph has drifted from its purpose or the sentence that states its point is hiding somewhere in the middle.

Editing tools: helpful, not a replacement

Grammar and style checkers catch typos and basic errors. They are useful, but they are no substitute for structural editing. AI writing tools can help you brainstorm phrasing, but you remain responsible for the accuracy and originality of every sentence, and you should check the policy of your faculty and of your target journal on their use. Never let a tool rewrite a technical claim without verifying that it still says exactly what you mean. No automated tool can fully carry out the three-pass check for you, because only you know what the paper is supposed to argue.

5.2

Giving and receiving peer feedback

Giving feedback that helps

Start with what is working. This is not mere politeness: it orients the writer and tells them what to keep. Be specific, because “this paragraph has two ideas” is more useful than “this is unclear”. Distinguish major issues of structure, logic and missing evidence from minor ones of wording. Where you can, ask questions instead of issuing commands; it respects the writer’s ownership of the text.

VAGUE AND UNACTIONABLE

“This section is confusing. Please fix it.”

SPECIFIC AND ACTIONABLE

“This paragraph moves between the sample and the outcome measure without a clear transition — consider splitting it into two paragraphs.”

Actionable feedback names the specific problem and points toward a specific fix, without rewriting the sentence for the author.

A STRUCTURE FOR WRITTEN FEEDBACK

1. Open with one or two genuine strengths.
2. Group the remaining comments by priority: structure first, then paragraphs, then sentences.
3. Limit yourself to the three to five most important issues.
4. Close by naming the single change that would most improve the draft.

Suppose a classmate's introduction has a strong opening but never states a gap or an aim. Useful feedback might read: "*Your opening makes the importance of the topic very clear. I could not find a sentence that says what is still unknown, or what your study set out to do — could you add both at the end of the introduction?*"

Receiving feedback without getting defensive

Read all the feedback once, fully, before reacting to any single comment. Assume good intent: a reviewer's job is to strengthen your paper, not to attack you. You need not agree with every comment, but you should understand why it was made; if a careful reader misunderstood a passage, other readers will too. When a comment is confusing, ask a clarifying question before guessing at its meaning.

5.3

Navigating journal peer review

Once a manuscript is submitted it follows a well-defined path.



TABLE 5.2 Common editorial decisions

Decision	What it means
Accept	Rare on a first submission; the paper needs no further changes
Minor revision	Small, clearly specified changes are requested
Major revision	Substantial changes are needed; the resubmission will be reviewed again
Reject	Not a fit for this journal, or fundamental issues were found

A request for major revision is good news. It means the journal is still interested and is telling you what would make the paper acceptable.

Writing a response to reviewers

Address every single comment, including those you disagree with; silence reads as avoidance. Structure the response point by point, quoting each comment before your reply. When you disagree, explain your reasoning respectfully and with evidence. Authors are allowed to push back. And state clearly what changed in the manuscript and where, for example "Added to Discussion, paragraph 2".

DEFENSIVE

"The reviewer clearly did not read our manuscript carefully. This point is already obvious."

PROFESSIONAL

"We thank the reviewer for this comment. We have clarified this point in the Methods (page 4, paragraph 2) to avoid ambiguity."

> Even a frustrating comment deserves a measured, constructive reply. Editors read these exchanges closely.

TABLE 5.3 Structure of a response letter

Element	Purpose
Reviewer comment (quoted)	Shows exactly what you are responding to
Your response	States agreement or disagreement, and your reasoning
Manuscript change	States exactly where and what was revised
Tone throughout	Thankful, concise and professional, even under criticism

Preprints

A preprint is a manuscript shared publicly before, or alongside, formal peer review. It speeds up the sharing of findings, invites early feedback and establishes priority of discovery. Because a preprint has not been peer reviewed, it is labelled as such and its conclusions should be treated with caution. Most major journals now allow preprints, but check your target journal's policy before posting.

Recognizing predatory journals

Not every publication that calls itself a journal operates as one. Three warning signs recur.

No real review
Accepts almost anything quickly, with little genuine peer review.

Fee-driven
Solicits submissions aggressively and charges high, mandatory fees.

Poor indexing
Not indexed in reputable databases such as PubMed or Scopus.

If an unfamiliar journal invites you to submit, ask your supervisor or a librarian before you reply.

5.4

Communicating beyond the paper

Patients, funders, journalists and policy makers rarely read the full paper. They read a summary, and many journals and funding bodies now require a plain-language summary alongside the abstract. Writing one is not "dumbing down". It is translating precisely, without jargon, for a different kind of expert: the reader. The exercise also shows you how well you understand your own findings.

TABLE 5.4 The same finding for two audiences

Technical abstract	Plain-language summary
"An 8-week aerobic exercise intervention significantly reduced systolic blood pressure (−8.2 mmHg, 95% CI: −11.4 to −5.0, p<0.001) in adults with stage 1 hypertension."	"Adults with mildly high blood pressure who walked briskly for 30 minutes, most days, for 8 weeks saw a meaningful drop in their blood pressure."
Assumes statistical literacy and field-specific vocabulary	Keeps the key finding; drops the statistics and jargon

The same principle applies to social media, where a growing number of researchers share their findings: translate precisely, and do not oversimplify. Never claim more certainty in a post than your evidence supports, because this is where overreach spreads fastest, and always link back to the original paper so that interested readers can check the full context.

Full paper	Abstract	Plain summary	Social post
Complete technical detail for expert peers.	Condensed technical summary for quick scanning.	The key finding, without jargon, for the public.	One clear sentence, linked to the source.

One finding, four audiences: the book ends where it began, with the question of who is reading.

A lifelong skill

Every skill in this book — structure, clarity, honest interpretation, careful revision — applies far beyond a single paper. You will use it in grant applications, in clinical documentation and in every manuscript you write. The habits of one idea per sentence, funnel-shaped introductions and objective results become faster with practice. They never become safe to skip.

KEY TAKEAWAYS

- Edit in passes: structure, then paragraphs, then sentences. Proofread last.
- Good feedback is specific, prioritised and kind; good authors read it without defensiveness.
- Answer every reviewer comment, point by point, and say exactly what you changed.
- A finding is not fully communicated until it has reached the readers who need it.

5.5

Practice

TRY IT

1. Reverse outline one page of your own writing: one line per paragraph, saying what it does. Is the order logical?
2. Exchange a paragraph with a classmate and give feedback using the four-step structure in section 5.2.
3. Write a two-sentence plain-language summary of a study you know. Give it to someone outside medicine and ask them to tell you what was found.

WRITING TASK 5

Does your opening sentence predict the paragraph?

Take the opening sentence of something you have written recently, for this course or elsewhere. Does it predict what the rest of the paragraph says? If not, rewrite it so that it does.

Check yourself: Structure — does each section do its IMRAD job, in order? Paragraph — does this paragraph have one clear topic sentence? Sentence — is it active, precise and non-redundant?

> *The best scientific writers are not the ones who get it right the first time — they are the ones who revise the most honestly.*

Review questions

5.1 Which editing pass should generally happen first?

A. Sentence-level grammar and word choice

B. Formatting the reference list

C. Structure and overall logical order

D. Checking figure resolution

5.2 The most useful peer feedback tends to be:

A. Specific, tied to a location in the text, and actionable

B. General praise, to keep morale up

C. A full rewrite of the author's sentences

D. Focused only on grammar and spelling

5.3 When you disagree with a reviewer's comment, the best approach is to:

A. Ignore the comment in your response

B. Withdraw the manuscript

C. Argue that the reviewer is unqualified

D. State your disagreement with clear reasoning and remain respectful

5.4 A plain-language summary should generally:

A. Include every statistic from the abstract

B. Preserve the key finding while removing jargon and statistics

C. Be written before the technical abstract

D. Replace the need for a full abstract

Appendix A · Checklists

These one-page checks gather the main tools of the book. Use them on any draft before you show it to someone else.

1 · DOES EACH SECTION DO ITS JOB?

- **Title** names the population, the intervention or exposure, and the outcome.
- **Abstract** can be understood without the paper and respects the journal's word limit.
- **Introduction** moves from context to what is known, to the gap, to the aim, and ends with the aim.
- **Methods** are in the past tense and detailed enough to repeat; they contain no results.
- **Results** give complete numbers in a logical order; they contain no interpretation.
- **Discussion** restates, compares, limits and signifies.

2 · SENTENCE AND PARAGRAPH CHECK

- Does each sentence make exactly one claim?
- Is active or passive voice chosen deliberately, not by habit?
- Has every redundant phrase been cut, and every vague quantifier replaced by a number?
- Does every paragraph open with its main point?
- Is each hedge matched to the strength of the evidence, with none stacked?

3 · RESULTS AND FIGURES CHECK

- Does every result give the value, its variability, the sample size and the test?
- Are null and negative results reported as fully as positive ones?
- Is the primary outcome reported first?
- Can each figure and table be understood from its caption alone?
- Do the axes represent the data honestly?

4 · BEFORE YOU SUBMIT

- Have all authors approved the final version, and does each meet the criteria for authorship?
- Are funding and conflicts of interest disclosed?
- Does every in-text citation appear in the reference list, in the journal's style?
- Is every paraphrase genuinely your own sentence, with its source cited?
- Have you done the three editing passes, and proofread last?

A RESPONSE TO REVIEWERS, IN OUTLINE

1. Thank the editor and reviewers, and summarize the main changes in two or three sentences.
2. For each comment: quote it, give your response, and state what changed and where.
3. Where you disagree, say so politely and give your evidence.
4. Keep the tone thankful, concise and professional throughout.

Appendix B · Answers to review questions

Chapter 1 · Foundations of Scientific Writing

1.1 **A.** Suspense — withholding the main finding until the end
Unlike a novel or news feature, a scientific paper tells the reader the main finding up front — in the abstract and often the introduction. Readers should never have to hunt for your conclusion.

1.2 **D.** A conference abstract
Conference abstracts are short, quickly reviewed, and designed exactly for preliminary or in-progress work — the format matches the stage of the research, not just its topic.

1.3 **C.** In the Discussion, before the conclusion
Limitations are an interpretive judgment about how much weight your findings should carry — that's Discussion work. Readers expect them just before your final conclusions.

1.4 **B.** On its own, without reading the rest of the paper
Databases, search engines, and busy readers often show only the abstract. If it depends on the full text to make sense, it has failed its main job.

Chapter 2 · Writing Clear and Accurate Sentences

2.1 **D.** Consider splitting it into separate sentences
Multiple 'and'-joined clauses often signal multiple ideas competing for the same sentence. Splitting lets each claim stand on its own and be verified independently.

2.2 **A.** The actor performing the action is obvious or unimportant
Passive voice earns its place when naming the actor would add nothing useful — e.g., routine lab procedures. It should never be used simply to sound more 'scientific.'

2.3 **B.** “This treatment may offer modest benefit, though further study is needed.”
It hedges exactly once, in proportion to weak-but-promising evidence — neither overclaiming a cure nor stacking redundant hedges.

2.4 **A.** “15 subjects were recruited.”
“A total of” and “for participation in the study” add no new information — the number and the verb “recruited” already say everything needed.

2.5 **C.** Adding a topic sentence and/or splitting it into focused paragraphs
A missing topic sentence and unrelated facts both point to the same root cause — the paragraph hasn't committed to one purpose yet. Give it one, or divide it into paragraphs that each can.

Chapter 3 · Building the Introduction and Literature Review

3.1 **B.** Is too broad to establish specific, concrete stakes
Sweeping historical openers delay the real content. Readers need concrete, current stakes within the first sentence or two — not a broad narrative windup.

3.2 **D.** Synthesize prior findings to set up your specific gap
A literature review is an argument in miniature: it builds, source by source, toward the specific gap your study will address — not a record of your reading list.

3.3 **A.** Trace the number back to the primary study and cite that
Numbers can drift or be misquoted as they pass through secondary sources. Always trace a specific statistic back to where it was originally measured and reported.

3.4 **C.** Map directly onto the gap identified just before it
The funnel only works if the narrowest point (the aim) directly answers the gap that came right before it. A mismatch here is one of the most common introduction weaknesses.

Chapter 4 · Presenting Results and Discussion

4.1 **A.** "This suggests that older patients respond better"
"Suggests" signals an inference about meaning or cause — that belongs in the Discussion. The other options simply report observed facts.

4.2 **C.** In whichever single format best serves the reader's need
Duplicating data across a figure and a table wastes space and reader attention. Choose the format — visual pattern or precise number lookup — that matches what the reader needs from that data.

4.3 **D.** "Because participants self-reported diet, recall bias may have affected accuracy."
It names the specific source of bias (self-report) and its likely effect (recall bias, accuracy) — precise enough that a reader can judge how much it matters.

4.4 **B.** State the finding's significance and plausible next steps
Readers should leave the Discussion understanding not just what you found and its limits, but why it matters and where the field might go next.

4.5 **D.** A form of self-plagiarism that should be avoided or properly disclosed
Journals consider undisclosed text reuse a form of self-plagiarism, even from your own prior work — publishers hold copyright, and readers assume new text is newly written.

Chapter 5 · Editing, Peer Review, and Beyond

5.1 **C.** Structure and overall logical order
Polishing sentences in a section that later gets deleted or moved wastes effort. Fix structure first, then paragraphs, then sentences — working top-down saves time.

5.2 **A.** Specific, tied to a location in the text, and actionable
Feedback that names a specific location and a specific issue lets the author actually act on it — vague praise or vague criticism both leave the writer guessing.

5.3 **D.** State your disagreement with clear reasoning and remain respectful
Editors expect authors to sometimes disagree with reviewers — what matters is that the disagreement is reasoned, evidence-based, and professionally worded.

5.4 **B.** Preserve the key finding while removing jargon and statistics
The goal is precise translation, not simplification that loses meaning — keep the true finding and its real-world implication, and let the technical detail go.

Appendix C · Glossary

Abstract
A self-contained summary of a paper, usually 200–300 words, that mirrors its structure in miniature.

Active voice
A construction in which the subject performs the action: "We measured blood pressure."

Confidence interval (CI)
A range of values that expresses the precision of an estimate; commonly reported at the 95% level.

Conflict of interest
A financial or personal interest that could bias, or appear to bias, the work.

Gap
The specific thing that remains unknown, untested or unresolved in the literature, and that a study sets out to address.

Hedge
A word such as "may" or "suggests" that signals the degree of confidence in a claim.

Hypothesis
A testable prediction that names the groups compared and the expected direction of an effect.

ICMJE
The International Committee of Medical Journal Editors, whose recommendations include widely used criteria for authorship.

IMRAD
Introduction, Methods, Results and Discussion: the standard structure of a research article.

Nominalization
A verb turned into an abstract noun, as when "investigate" becomes "conduct an investigation of".

Passive voice
A construction in which the subject receives the action: “Blood pressure was measured.”

Patchwriting
Replacing a few words in another author’s sentence while keeping its structure; a form of plagiarism.

Peer review
Evaluation of a manuscript by independent experts before a journal decides whether to publish it.

PICO
Population, Intervention (or exposure), Comparison, Outcome: a framework for structuring a clinical question.

Plain-language summary
A short account of a study’s key finding written without jargon for non-specialist readers.

Predatory journal
A publication that charges fees while providing little or no genuine peer review or editorial service.

Preprint
A manuscript shared publicly before, or alongside, formal peer review.

Primary source
An original research article reporting new data.

Reverse outline
An outline made from a finished draft by noting what each paragraph actually does.

Secondary source
A review article, textbook or other summary of work reported elsewhere.

Synthesis matrix
A table that sorts sources by theme so that agreement, conflict and gaps become visible.

Topic sentence
The sentence, usually the first, that states the single main point of a paragraph.

Appendix D · Further reading

The following sources are good next steps. The first three are free online and are the standards that journals themselves refer to.

International Committee of Medical Journal Editors. *Recommendations for the Conduct, Reporting, Editing, and Publication of Scholarly Work in Medical Journals.* Available at icmje.org.

EQUATOR Network. *Reporting guidelines for main study types* (CONSORT, STROBE, PRISMA, CARE and others). Available at equator-network.org.

Committee on Publication Ethics (COPE). *Core Practices and guidance on authorship, plagiarism and peer review.* Available at publicationethics.org.

Gopen GD, Swan JA. The science of scientific writing. *American Scientist.* 1990;78(6):550–558.

Schimel J. *Writing Science: How to Write Papers That Get Cited and Proposals That Get Funded.* Oxford University Press; 2012.

Zeiger M. *Essentials of Writing Biomedical Research Papers.* 2nd ed. McGraw-Hill; 2000.

Sword H. *Stylish Academic Writing.* Harvard University Press; 2012.

Swales JM, Feak CB. *Academic Writing for Graduate Students: Essential Tasks and Skills.* 3rd ed. University of Michigan Press; 2012.

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About ASIME

The Academic Society for International Medical Education (ASIME) is based in Nagoya, Japan. It supports international medical education through conferences, workshops, microcredentials and student experience programs. More information is available at asime.online.

> *Clear scientific writing does not depend on a large vocabulary or elegant style. It depends on structure, precision and honesty — and all three can be learned.*

A short, practical guide to writing research papers, built around five classes: the IMRAD structure, clear sentences, the introduction and literature review, results and discussion, and editing and peer review. With before-and-after examples, exercises, review questions and checklists.

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